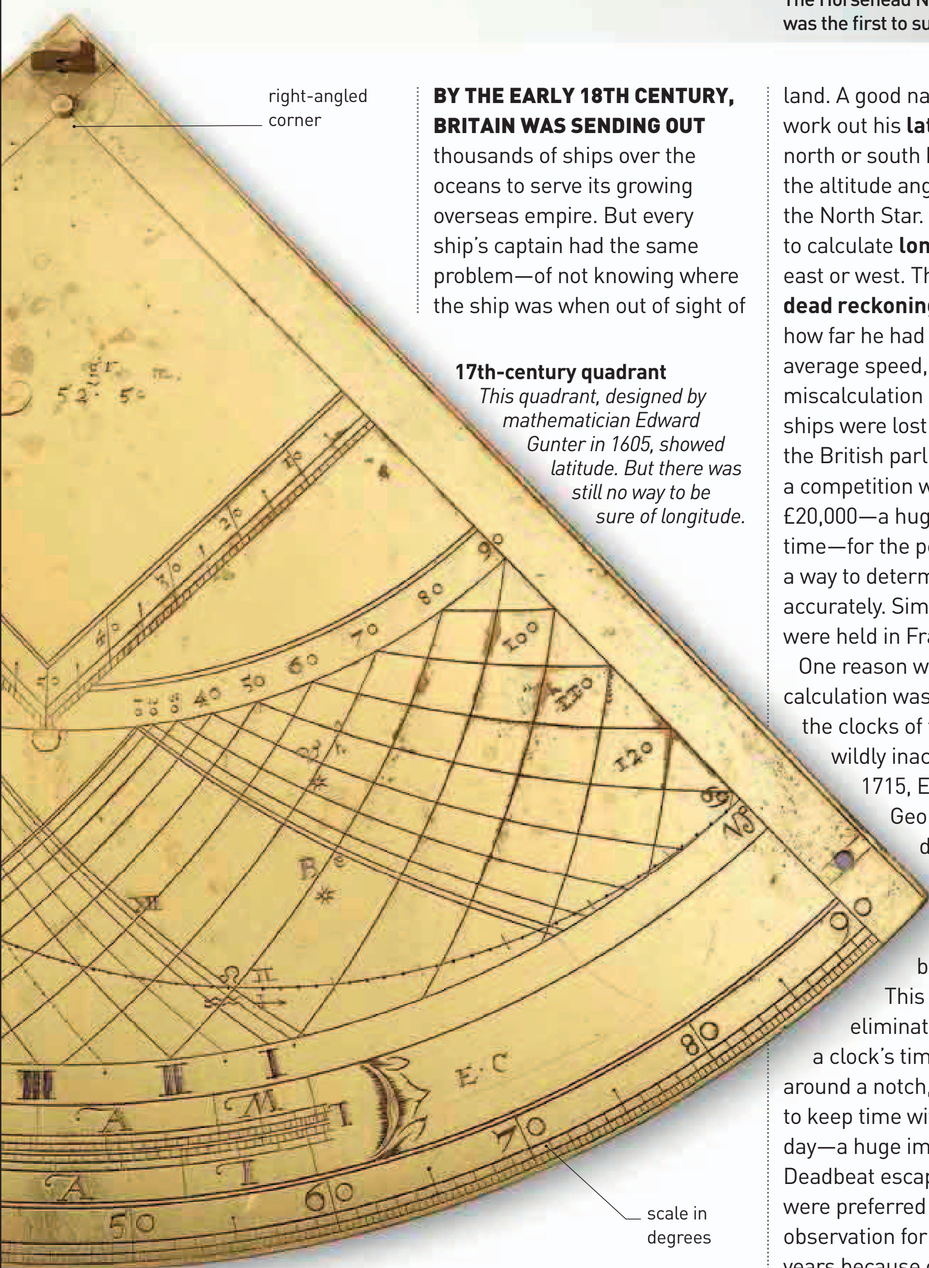
5 BILLION THE NUMBER OF YEARS BEFORE OUR SUN BECOMES A PLANETARY NEBULA

The Horsehead Nebula is a cloud of interstellar gas and dust. Edmond Halley was the first to suggest that indistinct objects in space could be nebulae.

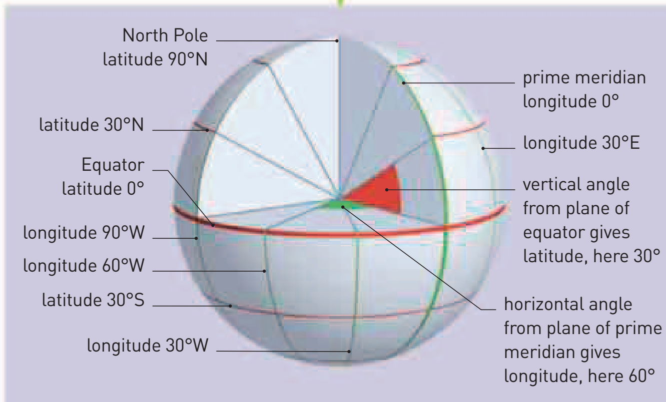
Giovanni Lancisi was first to realize that malaria is spread by mosquitos.



17th-century quadrant
This quadrant, designed by mathematician Edward Gunter in 1605, showed latitude. But there was still no way to be sure of longitude.

BY THE EARLY 18TH CENTURY, BRITAIN WAS SENDING OUT thousands of ships over the oceans to serve its growing overseas empire. But every ship's captain had the same problem—of not knowing where the ship was when out of sight of land. A good navigator could work out his **latitude**—how far north or south he was—from the altitude angle of the Sun and the North Star. The problem was to calculate **longitude**—how far east or west. The technique of **dead reckoning**, or estimating how far he had sailed from his average speed, gave a clue. But miscalculation meant that many ships were lost at sea. In 1714, the British parliament launched a competition with a prize of £20,000—a huge amount at the time—for the person who found a way to determine longitude accurately. Similar competitions were held in France and Holland.

One reason why longitude calculation was tricky was that the clocks of the day were wildly inaccurate. So, in 1715, English inventor George Graham's development of the **deadbeat escapement** was a great breakthrough. This mechanism eliminated recoil when a clock's time gear moved around a notch, enabling clocks to keep time within a second per day—a huge improvement. Deadbeat escapement clocks were preferred for scientific observation for the next 200 years because of their accuracy.



GEOGRAPHICAL COORDINATES

Solving the problem of how to find longitude at sea—how far east or west a ship is—was a priority in the 1700s. Lines of longitude, or meridians, run north-south around the world, dividing it like the segments of an orange. Zero degrees longitude is the Prime Meridian which passes through Greenwich, London, and a position's longitude is its angle east or west of this in degrees.

In 1715, English astronomer **Edmond Halley** suggested that the **age of Earth** could be determined by the **salinity of the oceans**, since the salt content would build up steadily as salt is washed in from the land. But his theory was impossible to prove, and, in fact, the salinity is too variable to be a measure of this. Halley was also the first astronomer to argue, correctly, that nebulae, which are seen as pale fuzzy shapes in the night sky, could comprise clouds of dust and gas.

IN 1716, ITALIAN PHYSICIAN GIOVANNI MARIA LANCISI (1654–1720) was the first to recognize the source of malaria. This often fatal disease, then common in Europe, was known as "ague" or "marsh fever" because it tended to occur near marshes, such as those around Rome. People believed it was caused by fumes from the damp ground—*mal'aria* is Italian for "bad air." But Lancisi realized that **malaria was caused by bites from swamp-inhabiting mosquitoes**. Few listened to

July 1714 British Parliament offers prize for a **method of determining longitude**

1715 Edmond Halley identifies five **nebulae**

1715 George Graham creates **deadbeat escapement** for clocks

Italian physician Giovanni Lansisi identifies **mosquitoes as carriers of malaria**